

Back-End Take-Home Challenge: Image Processing Service

Overview

Thank you for your interest in joining Ours Privacy's engineering team! This challenge is designed to assess your abilities in building developer-friendly APIs and services.

Task

Build a Cloudinary-style image processing service.

Requirements

- Start with an empty Git repository
- Create an API that can:
 - Receive image URL requests
 - Resize images to specified dimensions
 - Change image formatting (e.g., PNG to JPEG)
- **Error Handling:** Implement appropriate error handling and edge cases

Example API Requests

```
# Resize an image
GET /process?url=https://example.com/image.jpg&width=500&height=300

# Convert image format
GET /process?url=https://example.com/image.png&format=jpeg&qu
```

```
ality=80

# Combine operations
GET /process?url=https://example.com/image.jpg&width=800&height=600&format=webp&crop=fill

# Video thumbnail (bonus)
GET /video/thumbnail?url=https://example.com/video.mp4&time=15
```

Technical Guidelines

- **Language preference:** TypeScript (but you can choose your preferred language)
- **Documentation:** Include clear instructions on how to run and test your solution
- **API Design:** Focus on creating an intuitive, developer-friendly interface
- **Error Handling:** Implement appropriate error handling and edge cases

Evaluation Criteria

Your submission will be evaluated on:

1. **Core Functionality:** Completeness and correctness of the implementation
2. **Code Quality:** Organization, maintainability, and adherence to best practices
3. **Testing:** Coverage and effectiveness of tests
4. **API Design & User Experience:** Intuitiveness and documentation of interfaces. Ensure the user experience and capabilities the user can perform match what they would expect
5. **SDK Mindset:** Consideration for developer experience and API consumers' needs

Submission Instructions

- Please complete this challenge within 7 days
- Submit your code repository (GitHub, GitLab, etc.) link to your recruiter
- Include a README with setup instructions and any notes on your implementation

If you have any questions about the assignment, please don't hesitate to reach out to your recruiter.

We look forward to reviewing your solution!

Ours Privacy Engineering Team